English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Stochastic Empirical Methods, SE Year 3, 2023/2024 Bonus homework — *This assignment awards no points, but illustrates some types of argument required in the theory examination. Enjoy the work! Work on* $(\Omega, \mathcal{A}, \mathbb{P})$. *Write* $\overline{A} = \Omega \setminus A$ *and* $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

Problem 1. For $A, B \subset \Omega$, define $A \,\Delta\, B = (A \cap \overline{B}) \cup (\overline{A} \cap B)$. Express $\mathbb{1}_{(A \Delta B) \Delta C}$ using $\mathbb{1}_A, \mathbb{1}_B, \mathbb{1}_C$.

Problem 2. Prove for $A, B \subset \Omega$ with $\mathbb{P}(B) \neq 0$ that $\mathbb{P}(A \mid B) \geq [\mathbb{P}(A) + \mathbb{P}(B) - 1]/\mathbb{P}(B)$.

Problem 3. Pool samples from n people for a COVID test. If negative, declare everyone healthy; if positive, retest each individually. Four equally common blood groups each represent a quarter of the population, with infection proportions 10%, 1%, 5% and 4%, respectively. Each person's blood group is independent of the others, with these population proportions. The pooled test costs 1 BGN and each individual test 1.25 BGN.

- Model the expected cost $p_c(n)$ per tested person.
- Treat n as continuous x. Derive, without solving, an equation for x^* satisfying

$$p_c(x^*) = \min_{x>0} p_c(x).$$

- Optionally use a computer to find the optimal integer n^* and minimum cost.

Problem 4. Let X be $\mathbb{N}_0$-valued and $H_X(\lambda) = \mathbb{E}[e^{-\lambda X}]$ for $\lambda \geq 0$.

- Prove $H_X(\lambda) \leq 1$ and, by formally differentiating under the expectation, express $\mathbb{E}X$ and $\mathbb{E}X^2$.
- Express the generating function s_X using H_X.
- For $X \sim \mathrm{Bin}(n, p)$, find H_X and use it to obtain $\mathbb{E}X$, $\mathbb{E}X^2$.

Problem 5. A power plant's daily operation depends on $Y = |X - 1|$, the deviation of a parameter with $\mathbb{E}X = 1$, $DX = 0.01$. Because large deviations pose an environmental hazard, a regulator imposes a fine $g(n)$ BGN if $Y \in (n, n+1]$, $n \geq 2$.

- Find an upper bound for $f(a) = \mathbb{P}(Y > a)$.
- Independent experts claim that for $g(n) = n^{3/2}$, $\lim_{n\to\infty} g(n)\mathbb{P}(Y \in [n, n+1)) = 0$, making the fines insufficiently effective. Is their claim about the limit true?

Problem 6. Let $X_1, X_2, \ldots$ be independent Poisson variables with $\mathbb{P}(X_1 = k) = e^{-\lambda}\lambda^k/k!$, $k \geq 0$. Model a birth process by $Y_n = (1 + X_n)Y_{n-1}$, $Y_0 = 1$, and a birth-and-death process by $Z_n = X_n Z_{n-1}$, $Z_0 = 1$.

- Define almost sure convergence of random variables ξ_n to ξ.
- Does $n^{-1} \log Y_n$ have a limit, and what is it? What type of convergence holds? Does $\mathbb{E}Y_n$ grow exponentially?
- Calculate $\mathbb{P}(Z_n \neq 0)$ and prove that $Z_n \to 0$ almost surely, so extinction occurs almost surely. Does $\mathbb{E}Z_n$ grow exponentially when $\lambda > 1$?

Problem 7. Roll a six-sided die 3×10^{12} times and let X count ones or threes. A struggling student is asked to approximate:

- $\mathbb{P}(X > 10^{12})$;
- $\mathbb{P}(X > 10^{12} + 10^3)$;
- $\mathbb{P}(X > 10^{12} + 10^9)$.

The student answers "fifty-fifty", or 1/2, to all three. How many answers, and which, are correct? Justify mathematically.

Problem 8. The duration of acquired immunity is exponential X with rate λ. Immunity strength is a continuous variable Y with density $f_Y(y) = \beta y^{\beta-1}$, $0 < y \leq 1$, $\beta > 0$. Given n observations $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$, find maximum likelihood estimators of λ, β. Are they consistent?

Problem 9. In a phase 3 vaccine trial, $n = 10000$ people are tested and 9000 acquire immunity. Model the immune count as $X \sim \mathrm{Bin}(10000, p)$. Construct a symmetric critical region for $H_0 : p = 0.9$ with type I error $\alpha = 0.01$. Is there reason to regard this region as close to optimal?